

*4.10 CURVED MEMBERS

Our analysis of stresses due to bending has been restricted so far to members that are straight along their length. In this section, we will determine the bending stresses due to the application of equal and opposite couples to a member that is initially curved. Our discussion is limited to curved members with uniform cross sections possessing a plane of symmetry in which the bending couples are applied. It is assumed that all stresses remain below the proportional limit.

If the initial curvature of the member is small (i.e., the radius of curvature is large compared to the depth of its cross section) an approximation can be obtained for the distribution of stresses by assuming the member to be straight and using the formulas derived in Secs. 4.1B and 4.2.[†]

However, when the curvature is large, i.e., when the radius of curvature and the dimensions of the cross section of the member are of the same order of magnitude, it is necessary to use a different method of analysis, which was first introduced by the German engineer E. Winkler (1835–1888).

Consider the curved member of uniform cross section shown in Fig. 4.58. Its transverse section is symmetric with respect to the y axis (Fig. 4.58b) and, in its unstressed state, its upper and lower surfaces intersect the vertical xy plane along arcs of circle AB and FG centered at C (Fig. 4.58a). Now apply two equal

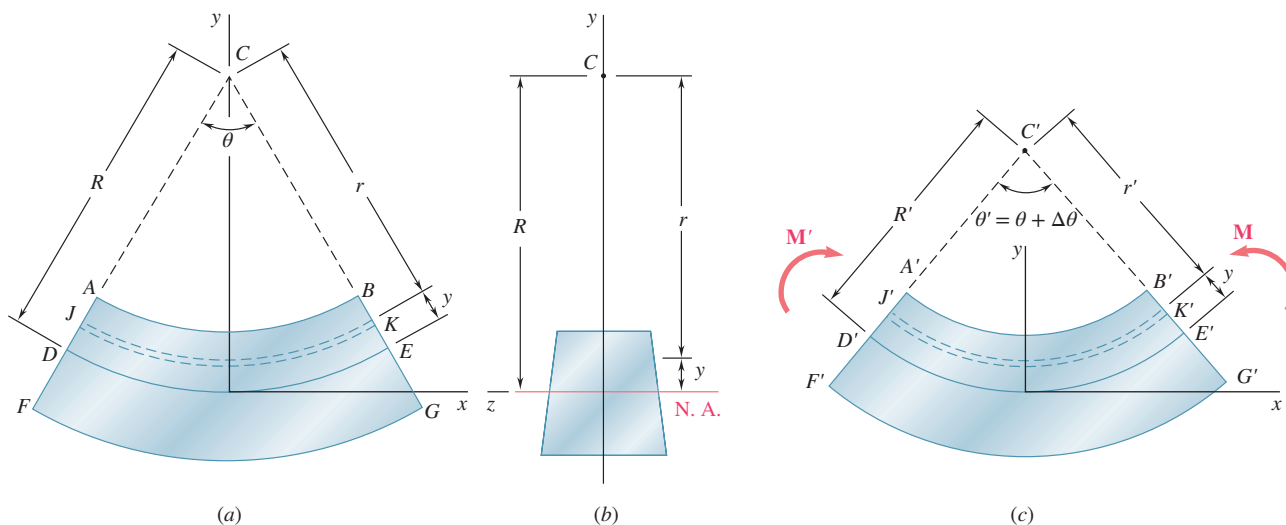


Fig. 4.58 Curved member in pure bending: (a) undeformed, (b) cross section, and (c) deformed.

and opposite couples $\mathbf{M}$ and $\mathbf{M}'$ in the plane of symmetry of the member (Fig. 4.58c). A reasoning similar to that of Sec. 4.1B would show that any transverse plane section containing C remains plane, and the various arcs of circle indicated in Fig. 4.58a are transformed into circular and concentric arcs with a center C' different from C . If the couples $\mathbf{M}$ and $\mathbf{M}'$ are directed as shown, the curvature of the various arcs of circle increases; that is, $A'C' < AC$. Also, the couples $\mathbf{M}$ and $\mathbf{M}'$ cause the length of the upper surface of the member to decrease ($A'B' < AB$) and the length of the lower surface to

[†]See Prob. 4.166.

increase ($F'G' > FG$). Therefore, we conclude that a *neutral surface* must exist in the member, the length of which remains constant. The intersection of the neutral surface with the xy plane is shown in Fig. 4.58a by the arc DE of radius R , and in Fig. 4.58c by the arc $D'E'$ of radius R' . The central angles θ and θ' corresponding respectively to DE and $D'E'$ express the fact that the length of the neutral surface remains constant by

$$R\theta = R'\theta' \quad (4.59)$$

Considering the arc of circle JK located at a distance y above the neutral surface and denoting respectively by r and r' the radius of this arc before and after the bending couples have been applied, the deformation of JK is

$$\delta = r'\theta' - r\theta \quad (4.60)$$

Observing from Fig. 4.58 that

$$r = R - y \quad r' = R' - y \quad (4.61)$$

and substituting these expressions into Eq. (4.60),

$$\delta = (R' - y)\theta' - (R - y)\theta$$

or recalling Eq. (4.59) and setting $\theta' - \theta = \Delta\theta$,

$$\delta = -y\Delta\theta \quad (4.62)$$

The normal strain ϵ_x in the elements of JK is obtained by dividing the deformation δ by the original length $r\theta$ of arc JK :

$$\epsilon_x = \frac{\delta}{r\theta} = -\frac{y\Delta\theta}{r\theta}$$

Recalling the first of the relationships in Eq. (4.61),

$$\epsilon_x = -\frac{\Delta\theta}{\theta} \frac{y}{R - y} \quad (4.63)$$

This relationship shows that, while each transverse section remains plane, the normal strain ϵ_x *does not vary linearly* with the distance y from the neutral surface.

The normal stress σ_x can be obtained from Hooke's law, $\sigma_x = E\epsilon_x$, by substituting for ϵ_x from Eq. (4.63):

$$\sigma_x = -\frac{E\Delta\theta}{\theta} \frac{y}{R - y} \quad (4.64)$$

or alternatively, recalling the first of Eqs. (4.61),

$$\sigma_x = -\frac{E\Delta\theta}{\theta} \frac{R - r}{r} \quad (4.65)$$

Equation (4.64) shows that, like ϵ_x , the normal stress σ_x *does not vary linearly* with the distance y from the neutral surface. Plotting σ_x versus y , an arc of hyperbola is obtained (Fig. 4.59).

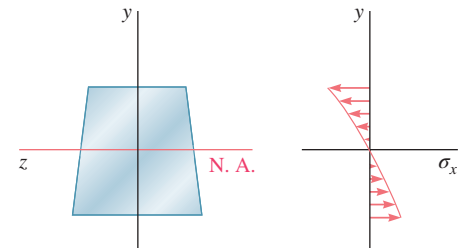


Fig. 4.59 Stress distribution in curved beam.

To determine the location of the neutral surface in the member and the value of the coefficient $E \Delta\theta/\theta$ used in Eqs. (4.64) and (4.65), we recall that the elementary forces acting on any transverse section must be statically equivalent to the bending couple $\mathbf{M}$. Expressing that the sum of the elementary forces acting on the section must be zero and that the sum of their moments about the transverse z axis must be equal to the bending moment M , write the equations

$$\int \sigma_x dA = 0 \quad (4.1)$$

and

$$\int (-y\sigma_x dA) = M \quad (4.3)$$

Substituting for σ_x from Eq. (4.65) into Eq. (4.1), write

$$-\int \frac{E \Delta\theta}{\theta} \frac{R-r}{r} dA = 0$$

$$\int \frac{R-r}{r} dA = 0$$

$$R \int \frac{dA}{r} - \int dA = 0$$

from which it follows that the distance R from the center of curvature C to the neutral surface is defined by

$$R = \frac{A}{\int \frac{dA}{r}} \quad (4.66)$$

Note that the value obtained for R is not equal to the distance $\bar{r}$ from C to the centroid of the cross section, since $\bar{r}$ is defined by a different relationship, namely,

$$\bar{r} = \frac{1}{A} \int r dA \quad (4.67)$$

Thus in a curved member *the neutral axis of a transverse section does not pass through the centroid of that section* (Fig. 4.60).[†] Expressions for the

[†]However, an interesting property of the neutral surface is noted if Eq. (4.66) is written in the alternative form

$$\frac{1}{R} = \frac{1}{A} \int \frac{1}{r} dA \quad (4.66a)$$

Equation (4.66a) shows that, if the member is divided into a large number of fibers of cross-sectional area dA , the curvature $1/R$ of the neutral surface is equal to the average value of the curvature $1/r$ of the various fibers.

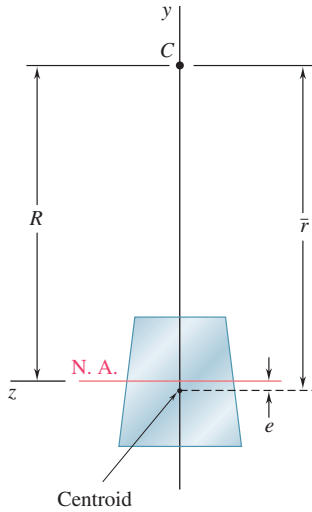


Fig. 4.60 Parameter e locates neutral axis relative to the centroid of a curved member section.

radius R of the neutral surface will be derived for some specific cross-sectional shapes in Concept Application 4.10 and in Probs. 4.187 through 4.189. These expressions are shown in Fig. 4.61.

Substituting now for σ_x from Eq. (4.65) into Eq. (4.3), write

$$\int \frac{E \Delta \theta}{\theta} \frac{R - r}{r} y dA = M$$

or since $y = R - r$,

$$\frac{E \Delta \theta}{\theta} \int \frac{(R - r)^2}{r} dA = M$$

Expanding the square in the integrand, we obtain after reductions

$$\frac{E \Delta \theta}{\theta} \left[R^2 \int \frac{dA}{r} - 2RA + \int r dA \right] = M$$

Recalling Eqs. (4.66) and (4.67), we note that the first term in the brackets is equal to RA , while the last term is equal to $\bar{r}A$. Therefore,

$$\frac{E \Delta \theta}{\theta} (RA - 2RA + \bar{r}A) = M$$

and solving for $E \Delta \theta / \theta$,

$$\frac{E \Delta \theta}{\theta} = \frac{M}{A(\bar{r} - R)} \quad (4.68)$$

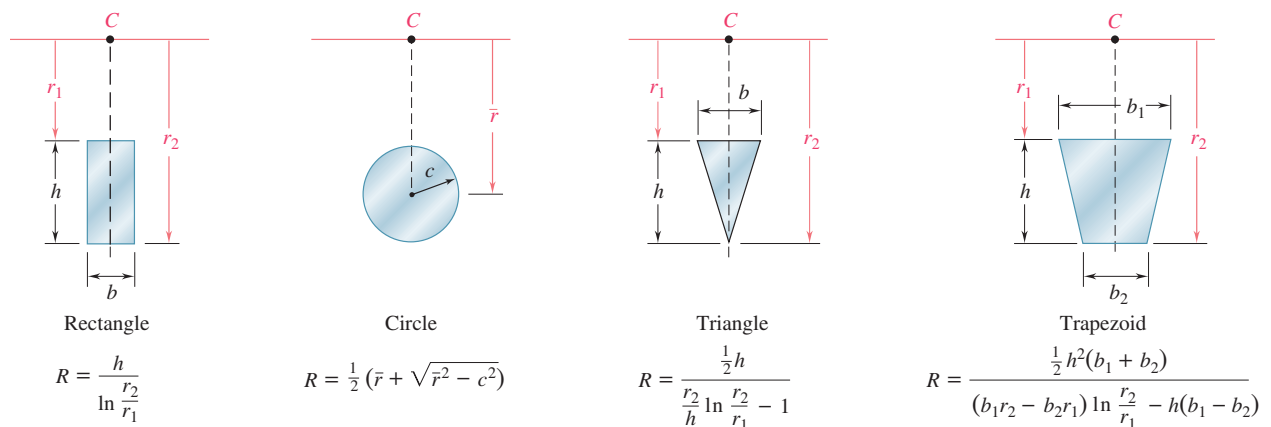


Fig. 4.61 Radius of neutral surface for various cross-sectional shapes.

Referring to Fig. 4.58, $\Delta\theta > 0$ for $M > 0$. It follows that $\bar{r} - R > 0$, or $R < \bar{r}$, regardless of the shape of the section. Thus the neutral axis of a transverse section is always located between the centroid of the section and the center of curvature of the member (Fig. 4.60). Setting $\bar{r} - R = e$, Eq. (4.68) is written in the form

$$\frac{E \Delta\theta}{\theta} = \frac{M}{Ae} \quad (4.69)$$

Substituting for $E \Delta\theta/\theta$ from Eq. (4.69) into Eqs. (4.64) and (4.65), the alternative expressions for the normal stress σ_x in a curved beam are

$$\sigma_x = -\frac{My}{Ae(R - y)} \quad (4.70)$$

and

$$\sigma_x = \frac{M(r - R)}{Aer} \quad (4.71)$$

Note that the parameter e in the previous equations is a small quantity obtained by subtracting two lengths of comparable size, R and $\bar{r}$. To determine σ_x with a reasonable degree of accuracy, it is necessary to compute R and $\bar{r}$ very accurately, particularly when both of these quantities are large (i.e., when the curvature of the member is small). However, it is possible in such a case to obtain a good approximation for σ_x by using the formula $\sigma_x = -My/I$ developed for straight members.

We will now determine the change in curvature of the neutral surface caused by the bending moment M . Solving Eq. (4.59) for the curvature $1/R'$ of the neutral surface in the deformed member,

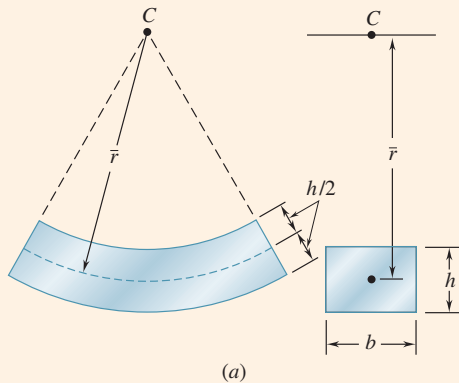
$$\frac{1}{R'} = \frac{1}{R} \frac{\theta'}{\theta}$$

or setting $\theta' = \theta + \Delta\theta$ and recalling Eq. (4.69),

$$\frac{1}{R'} = \frac{1}{R} \left(1 + \frac{\Delta\theta}{\theta} \right) = \frac{1}{R} \left(1 + \frac{M}{EAe} \right)$$

the change in curvature of the neutral surface is

$$\frac{1}{R'} - \frac{1}{R} = \frac{M}{EAeR} \quad (4.72)$$



Concept Application 4.10

A curved rectangular bar has a mean radius $\bar{r} = 6$ in. and a cross section of width $b = 2.5$ in. and depth $h = 1.5$ in. (Fig. 4.62a). Determine the distance e between the centroid and the neutral axis of the cross section.

We first derive the expression for the radius R of the neutral surface. Denoting by r_1 and r_2 , respectively, the inner and outer radius of the bar (Fig. 4.62b), use Eq. (4.66) to write

$$R = \frac{A}{\int_{r_1}^{r_2} \frac{dA}{r}} = \frac{bh}{\int_{r_1}^{r_2} \frac{dr}{r}} = \frac{h}{\ln \frac{r_2}{r_1}} \quad (4.73)$$

For the given data,

$$r_1 = \bar{r} - \frac{1}{2}h = 6 - 0.75 = 5.25 \text{ in.}$$

$$r_2 = \bar{r} + \frac{1}{2}h = 6 + 0.75 = 6.75 \text{ in.}$$

Substituting for h , r_1 , and r_2 into Eq. (4.73),

$$R = \frac{h}{\ln \frac{r_2}{r_1}} = \frac{1.5 \text{ in.}}{\ln \frac{6.75}{5.25}} = 5.9686 \text{ in.}$$

The distance between the centroid and the neutral axis of the cross section (Fig. 4.62c) is thus

$$e = \bar{r} - R = 6 - 5.9686 = 0.0314 \text{ in.}$$

Note that it was necessary to calculate R with five significant figures in order to obtain e with the usual degree of accuracy.

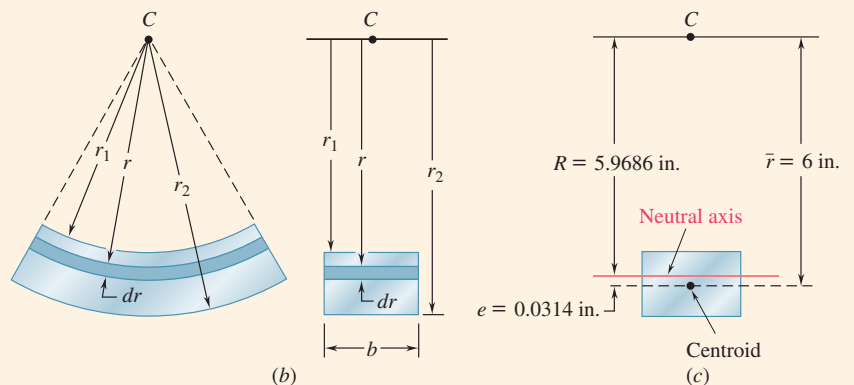


Fig. 4.62 (a) Curved rectangular bar. (b) Dimensions for curved bar. (c) Location of the neutral axis.

Concept Application 4.11

For the bar of Concept Application 4.10, determine the largest tensile and compressive stresses, knowing that the bending moment in the bar is $M = 8 \text{ kip}\cdot\text{in.}$

Use Eq. (4.71) with the given data

$$M = 8 \text{ kip}\cdot\text{in.} \quad A = bh = (2.5 \text{ in.})(1.5 \text{ in.}) = 3.75 \text{ in}^2$$

and the values obtained in Concept Application 4.10 for R and e :

$$R = 5.969 \quad e = 0.0314 \text{ in.}$$

First using $r = r_2 = 6.75 \text{ in.}$ in Eq. (4.71), write

$$\begin{aligned} \sigma_{\max} &= \frac{M(r_2 - R)}{Aer_2} \\ &= \frac{(8 \text{ kip}\cdot\text{in.})(6.75 \text{ in.} - 5.969 \text{ in.})}{(3.75 \text{ in}^2)(0.0314 \text{ in.})(6.75 \text{ in.})} \\ &= 7.86 \text{ ksi} \end{aligned}$$

Now using $r = r_1 = 5.25 \text{ in.}$ in Eq. (4.71),

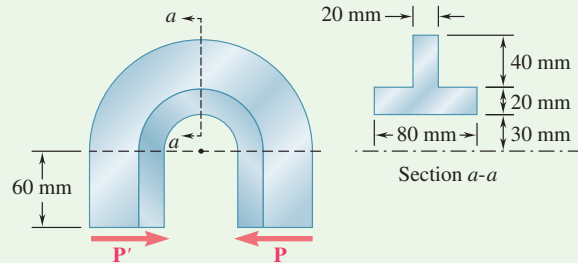
$$\begin{aligned} \sigma_{\min} &= \frac{M(r_1 - R)}{Aer_1} \\ &= \frac{(8 \text{ kip}\cdot\text{in.})(5.25 \text{ in.} - 5.969 \text{ in.})}{(3.75 \text{ in}^2)(0.0314 \text{ in.})(5.25 \text{ in.})} \\ &= -9.30 \text{ ksi} \end{aligned}$$

Remark. Compare the values obtained for $\sigma_{\max}$ and $\sigma_{\min}$ with the result for a straight bar. Using Eq. (4.15) of Sec. 4.2,

$$\begin{aligned} \sigma_{\max, \min} &= \pm \frac{Mc}{I} \\ &= \pm \frac{(8 \text{ kip}\cdot\text{in.})(0.75 \text{ in.})}{\frac{1}{12}(2.5 \text{ in.})(1.5 \text{ in.})^3} = \pm 8.53 \text{ ksi} \end{aligned}$$

Sample Problem 4.11

A machine component has a T-shaped cross section and is loaded as shown. Knowing that the allowable compressive stress is 50 MPa, determine the largest force P that can be applied to the component.



STRATEGY: The properties are first determined for the singly symmetric cross section. The force and couple at the critical section are used to calculate the maximum compressive stress, which is obtained by superposing the axial stress and the bending stress determined from Eqs. (4.66) and (4.71). This stress is then equated to the allowable compressive stress to determine the force P .

MODELING and ANALYSIS:

Centroid of the Cross Section. Locate the centroid D of the cross section (Fig. 1).

	A_i, mm^2	$\bar{r}_i, \text{mm}$	$\bar{r}_i A_i, \text{mm}^3$	$\bar{r} \Sigma A_i = \Sigma \bar{r}_i A_i$
1	$(20)(80) = 1600$	40	64×10^3	$\bar{r}(2400) = 120 \times 10^3$
2	$(40)(20) = 800$	70	56×10^3	$\bar{r} = 50 \text{ mm} = 0.050 \text{ m}$
	$\Sigma A_i = 2400$		$\Sigma \bar{r}_i A_i = 120 \times 10^3$	

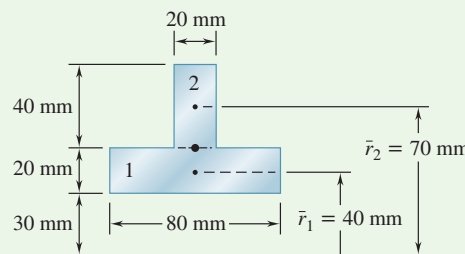


Fig. 1 Composite areas to calculate centroid location.

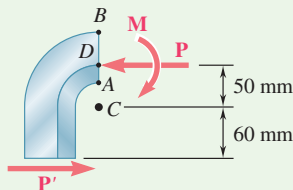


Fig. 2 Free-body diagram of left side.

Force and Couple at D . The internal forces in section $a-a$ are equivalent to a force P acting at D and a couple M of moment (Fig. 2)

$$M = P(50 \text{ mm} + 60 \text{ mm}) = (0.110 \text{ m})P$$

(continued)

Superposition. The centric force $\mathbf{P}$ causes a uniform compressive stress on section $a-a$, shown in Fig. 3a. The bending couple $\mathbf{M}$ causes a varying stress distribution [Eq. (4.71)], shown in Fig. 3b. We note that the couple $\mathbf{M}$ tends to increase the curvature of the member and is therefore positive (see Fig. 4.58). The total stress at a point of section $a-a$ located at distance r from the center of curvature C is

$$\sigma = -\frac{P}{A} + \frac{M(r-R)}{Aer} \quad (1)$$

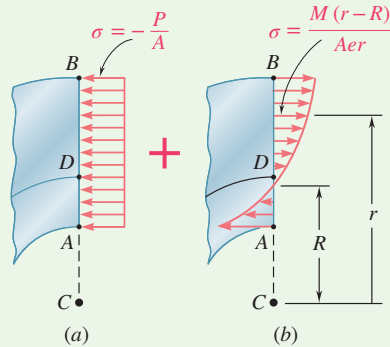


Fig. 3 Stress distribution is the superposition of (a) axial stress and (b) bending stress.

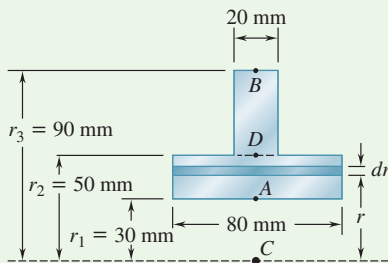


Fig. 4 Geometry of cross section.

Radius of Neutral Surface. Using Fig. 4, we now determine the radius R of the neutral surface by using Eq. (4.66).

$$\begin{aligned} R &= \frac{A}{\int \frac{dA}{r}} = \frac{2400 \text{ mm}^2}{\int_{r_1}^{r_2} \frac{(80 \text{ mm}) dr}{r} + \int_{r_2}^{r_3} \frac{(20 \text{ mm}) dr}{r}} \\ &= \frac{2400}{80 \ln \frac{50}{30} + 20 \ln \frac{90}{50}} = \frac{2400}{40.866 + 11.756} = 45.61 \text{ mm} \\ &= 0.04561 \text{ m} \end{aligned}$$

We also compute: $e = \bar{r} - R = 0.05000 \text{ m} - 0.04561 \text{ m} = 0.00439 \text{ m}$

Allowable Load. We observe that the largest compressive stress will occur at point A where $r = 0.030 \text{ m}$. Recalling that $\sigma_{\text{all}} = 50 \text{ MPa}$ and using Eq. (1), write

$$-50 \times 10^6 \text{ Pa} = -\frac{P}{2.4 \times 10^{-3} \text{ m}^2} + \frac{(0.110 P)(0.030 \text{ m} - 0.04561 \text{ m})}{(2.4 \times 10^{-3} \text{ m}^2)(0.00439 \text{ m})(0.030 \text{ m})}$$

$$-50 \times 10^6 = -417P - 5432P$$

$$P = 8.55 \text{ kN} \quad \blacktriangleleft$$

Problems

For the curved bar and loading shown, determine the stress at point A when (a) $r_1 = 30$ mm, (b) $r_1 = 50$ mm.

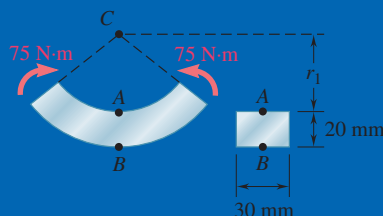


Fig. P4.161 and P4.162

For the curved bar and loading shown, determine the stress at points A and B when $r_1 = 40$ mm.

For the machine component and loading shown, determine the stress at point A when (a) $h = 2$ in., (b) $h = 2.6$ in.

For the machine component and loading shown, determine the stress at points A and B when $h = 2.5$ in.

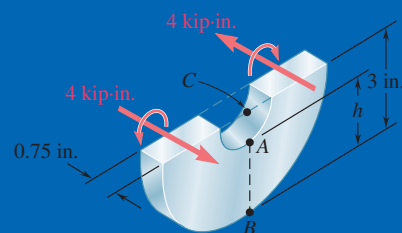


Fig. P4.163 and P4.164

- 4.165** The curved bar shown has a cross section of 40×60 mm and an inner radius $r_1 = 15$ mm. For the loading shown, determine the largest tensile and compressive stresses.

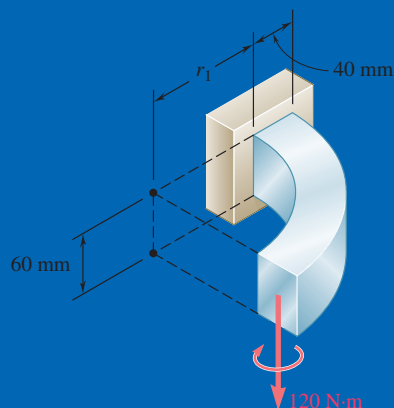


Fig. P4.165 and P4.166

- 4.166** For the curved bar and loading shown, determine the percent error introduced in the computation of the maximum stress by assuming that the bar is straight. Consider the case when (a) $r_1 = 20$ mm, (b) $r_1 = 200$ mm, (c) $r_1 = 2$ m.

- 4.167** Steel links having the cross section shown are available with different central angles β . Knowing that the allowable stress is 12 ksi, determine the largest force P that can be applied to a link for which $\beta = 90^\circ$.

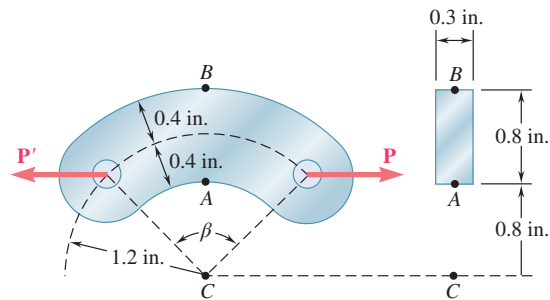


Fig. P4.167

- 4.168** Solve Prob. 4.167, assuming that $\beta = 60^\circ$.

- 4.169** The curved bar shown has a cross section of 30×30 mm. Knowing that the allowable compressive stress is 175 MPa, determine the largest allowable distance a .

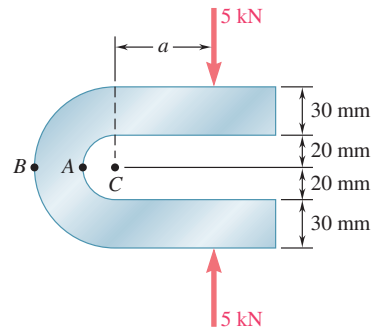


Fig. P4.169

- 4.170** For the split ring shown, determine the stress at (a) point A, (b) point B.

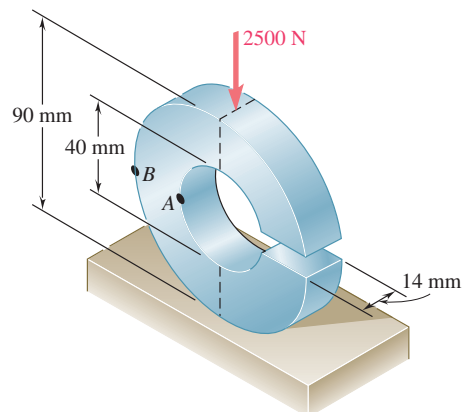


Fig. P4.170

- 4.171** Three plates are welded together to form the curved beam shown. For $M = 8 \text{ kip}\cdot\text{in.}$, determine the stress at (a) point A, (b) point B, (c) the centroid of the cross section.

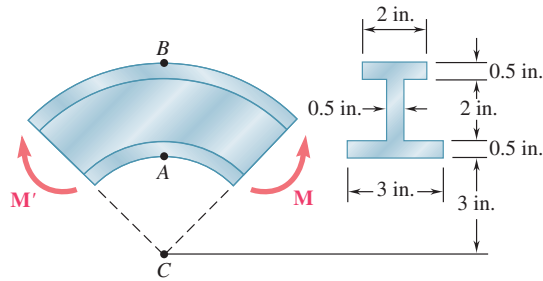


Fig. P4.171 and P4.172

- 4.172** Three plates are welded together to form the curved beam shown. For the given loading, determine the distance e between the neutral axis and the centroid of the cross section.
- 4.173** A machine component has a T-shaped cross section that is orientated as shown. Knowing that $M = 2.5 \text{ kN}\cdot\text{m}$, determine the stress at (a) point A, (b) point B.

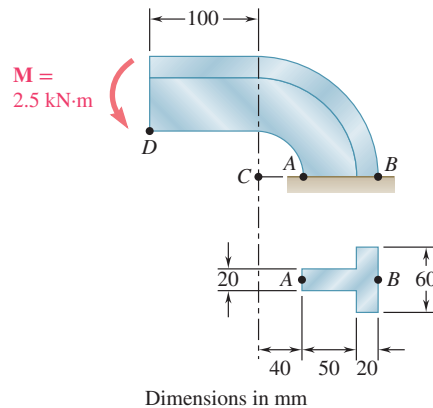


Fig. P4.173 and P4.174

- 4.174** Assuming that the couple shown is replaced by a vertical 10-kN force attached at point D and acting downward, determine the stress at (a) point A, (b) point B.
- 4.175** The split ring shown has an inner radius $r_1 = 0.8 \text{ in.}$ and a circular cross section of diameter $d = 0.6 \text{ in.}$ Knowing that each of the 120-lb forces is applied at the centroid of the cross section, determine the stress (a) at point A, (b) at point B.
- 4.176** Solve Prob. 4.175, assuming that the ring has an inner radius $r_1 = 0.6 \text{ in.}$ and a cross-sectional diameter $d = 0.8 \text{ in.}$

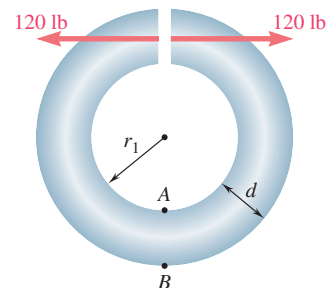


Fig. P4.175

- 4.177** The bar shown has a circular cross section of 14-mm diameter. Knowing that $a = 32$ mm, determine the stress at (a) point A, (b) point B.

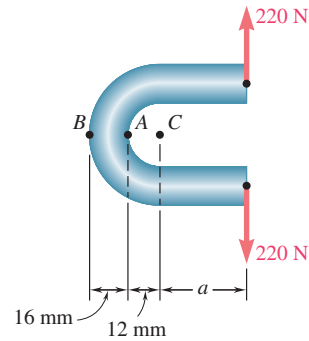


Fig. P4.177 and P4.178

- 4.178** The bar shown has a circular cross section of 14-mm diameter. Knowing that the allowable stress is 38 MPa, determine the largest permissible distance a from the line of action of the 220-N forces to the plane containing the center of curvature of the bar.

- 4.179** The curved bar shown has a circular cross section of 32-mm diameter. Determine the largest couple M that can be applied to the bar about a horizontal axis if the maximum stress is not to exceed 60 MPa.

- 4.180** Knowing that $P = 10$ kN, determine the stress at (a) point A, (b) point B.

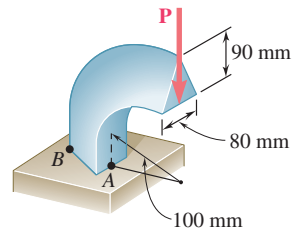


Fig. P4.180

- 4.181 and 4.182** Knowing that $M = 5$ kip-in., determine the stress at (a) point A, (b) point B.

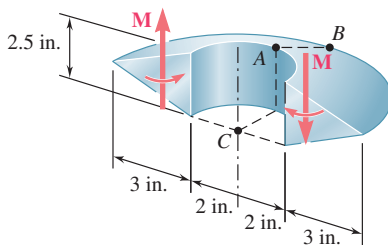


Fig. P4.181

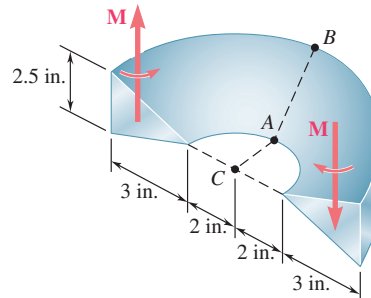


Fig. P4.182

- 4.183** Knowing that the machine component shown has a trapezoidal cross section with $a = 3.5$ in. and $b = 2.5$ in., determine the stress at (a) point A, (b) point B.

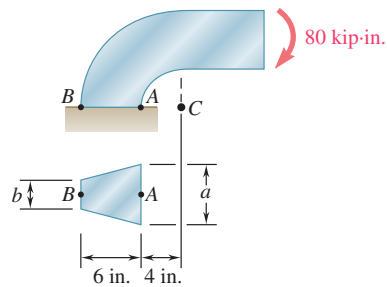


Fig. P4.183 and P4.184

- 4.184** Knowing that the machine component shown has a trapezoidal cross section with $a = 2.5$ in. and $b = 3.5$ in., determine the stress at (a) point A, (b) point B.

- 4.185** For the curved beam and loading shown, determine the stress at (a) point A, (b) point B.

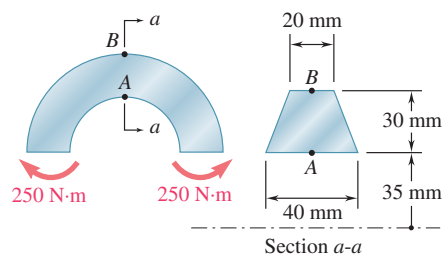


Fig. P4.185

- 4.186** For the crane hook shown, determine the largest tensile stress in section $a-a$.

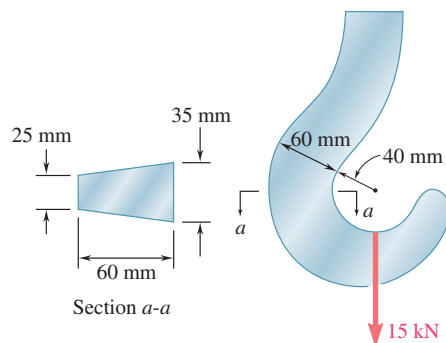


Fig. P4.186

- *4.187 through 4.189** Using Eq. (4.66), derive the expression for R given in Fig. 4.61 for
- *4.187** A circular cross section.
 - 4.188** A trapezoidal cross section.
 - 4.189** A triangular cross section.

- 4.190** Show that if the cross section of a curved beam consists of two or more rectangles, the radius R of the neutral surface can be expressed as

$$R = \frac{A}{\ln \left[\left(\frac{r_2}{r_1} \right)^{b_1} \left(\frac{r_3}{r_2} \right)^{b_2} \left(\frac{r_4}{r_3} \right)^{b_3} \right]}$$

where A is the total area of the cross section.

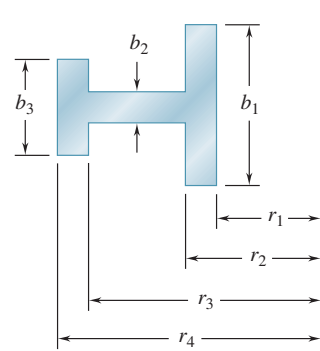


Fig. P4.190

- *4.191** For a curved bar of rectangular cross section subjected to a bending couple $\mathbf{M}$, show that the radial stress at the neutral surface is

$$\sigma_r = \frac{M}{Ae} \left(1 - \frac{r_1}{R} - \ln \frac{R}{r_1} \right)$$

and compute the value of σ_r for the curved bar of Concept Applications 4.10 and 4.11. (*Hint:* Consider the free-body diagram of the portion of the beam located above the neutral surface.)

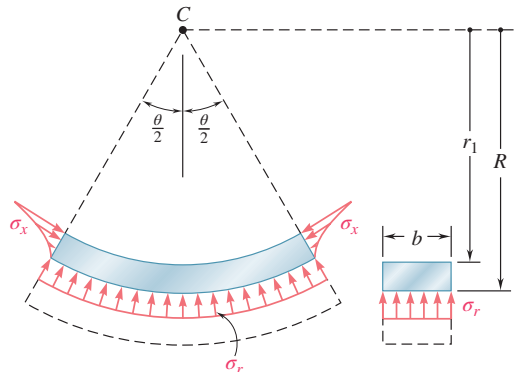


Fig. P4.191

Review and Summary

This chapter was devoted to the analysis of members in *pure bending*. The stresses and deformation in members subjected to equal and opposite couples $\mathbf{M}$ and $\mathbf{M}'$ acting in the same longitudinal plane (Fig. 4.63) were studied.

In members possessing a plane of symmetry and subjected to couples acting in that plane, it was proven that *transverse sections remain plane* as a member is deformed. A member in pure bending also has a *neutral surface* along which normal strains and stresses are zero. The longitudinal *normal strain* ϵ_x *varies linearly* with the distance y from the neutral surface:

$$\epsilon_x = -\frac{y}{\rho} \quad (4.8)$$

where ρ is the *radius of curvature* of the neutral surface (Fig. 4.64). The intersection of the neutral surface with a transverse section is known as the *neutral axis* of the section.

For members made of a material that follows Hooke's law, the *normal stress* σ_x *varies linearly* with the distance from the neutral axis (Fig. 4.65). Using the maximum stress σ_m , the normal stress is

$$\sigma_x = -\frac{y}{c} \sigma_m \quad (4.12)$$

where c is the largest distance from the neutral axis to a point in the section.

By setting the sum of the elementary forces $\sigma_x dA$ equal to zero, we proved that the *neutral axis passes through the centroid* of the cross section of a member in pure bending. Then by setting the sum of the moments of the elementary forces equal to the bending moment, the *elastic flexure formula* is

$$\sigma_m = \frac{Mc}{I} \quad (4.15)$$

where I is the moment of inertia of the cross section with respect to the neutral axis. The normal stress at any distance y from the neutral axis is

$$\sigma_x = -\frac{My}{I} \quad (4.16)$$

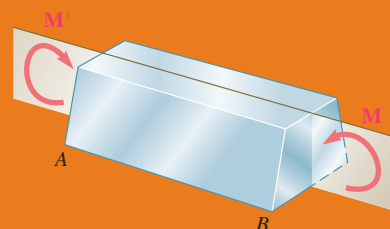


Fig. 4.63 Member in pure bending.

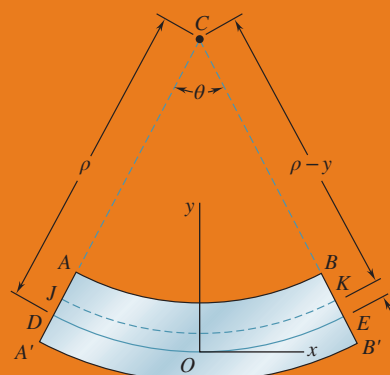


Fig. 4.64 Deformation with respect to neutral axis.

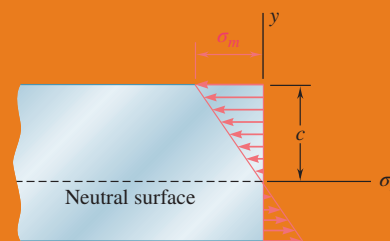


Fig. 4.65 Stress distribution for the elastic flexure formula.

Noting that I and c depend only on the geometry of the cross section we introduced the *elastic section modulus*

$$S = \frac{I}{c} \quad (4.17)$$

Use the section modulus to write an alternative expression for the maximum normal stress:

$$\sigma_m = \frac{M}{S} \quad (4.18)$$

The *curvature* of a member is the reciprocal of its radius of curvature, and may be found by

$$\frac{1}{\rho} = \frac{M}{EI} \quad (4.21)$$

In the bending of homogeneous members possessing a plane of symmetry, deformations occur in the plane of a transverse cross section and result in *anticlastic curvature* of the members.

We considered the bending of members made of several materials with *different moduli of elasticity*. While transverse sections remain plane, the *neutral axis does not pass through the centroid* of the composite cross section (Fig. 4.66). Using the ratio of the moduli of elasticity of the materials, we obtained a *transformed section* corresponding to an equivalent member made entirely of one material. The methods previously developed are used to determine the stresses in this equivalent homogeneous member (Fig. 4.67), and the ratio of the moduli of elasticity is used to determine the stresses in the composite beam.

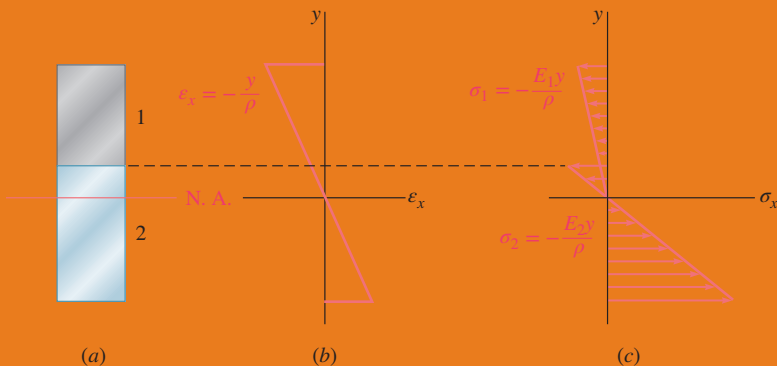


Fig. 4.66 (a) Composite section. (b) Strain distribution. (c) Stress distribution.

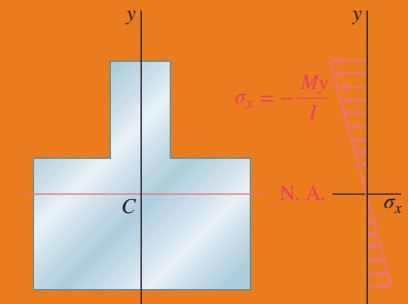


Fig. 4.67 Transformed section.

Stress concentrations occur in members in pure bending and were discussed; charts giving stress-concentration factors for flat bars with fillets and grooves also were presented in Figs. 4.24 and 4.25.

A rectangular beam made of an *elastoplastic material* (Fig. 4.68) was analyzed as the magnitude of the bending moment was increased (Fig. 4.69). The *maximum elastic moment* M_Y occurs when yielding is initiated in the beam (Fig. 4.69b). As the bending moment is increased, plastic zones develop (Fig. 4.69c), and the size of the elastic core of the member is decreased. When the beam becomes fully plastic (Fig. 4.69d), the maximum or *plastic moment* M_p is obtained. *Permanent deformations* and *residual stresses* remain in a member after the loads that caused yielding have been removed.

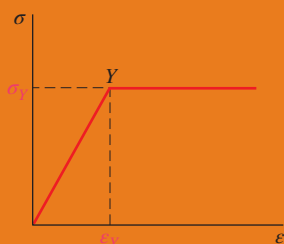


Fig. 4.68 Elastoplastic stress-strain diagram.

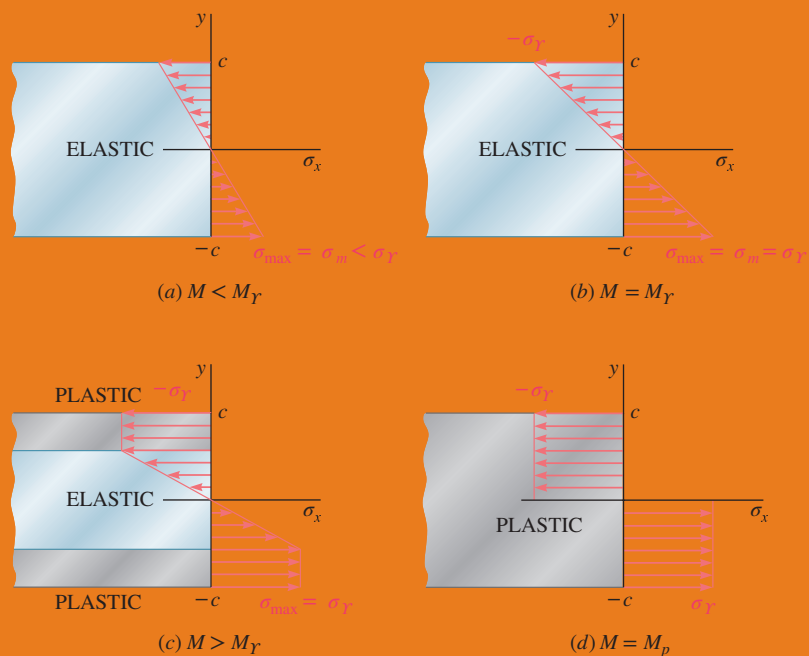


Fig. 4.69 Bending stress distribution in a member for (a) elastic, $M < M_Y$; (b) yield impending, $M = M_Y$; (c) partially yielded, $M > M_Y$; and (d) fully plastic, $M = M_p$.

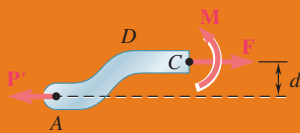


Fig. 4.70 Section of an eccentrically loaded member.

When a member is loaded *eccentrically in a plane of symmetry*, the *eccentric load* is replaced with a force-couple system located at the centroid of the cross section (Fig. 4.70). The stresses from the centric load and the bending couple are superposed (Fig. 4.71):

$$\sigma_x = \frac{P}{A} - \frac{My}{I} \quad (4.50)$$

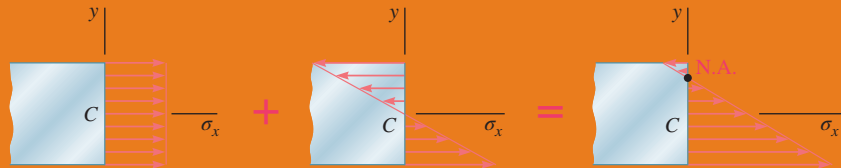


Fig. 4.71 Stress distribution for eccentric loading is obtained by superposing the axial and pure bending distributions.

For bending of members of *unsymmetric cross section*, the flexure formula may be used, provided that the couple vector $\mathbf{M}$ is directed along one of the principal centroidal axes of the cross section. When necessary, $\mathbf{M}$ can be resolved into components along the principal axes, and the stresses superposed due to the component couples (Figs. 4.72 and 4.73).

$$\sigma_x = -\frac{M_z y}{I_z} + \frac{M_y z}{I_y} \quad (4.55)$$

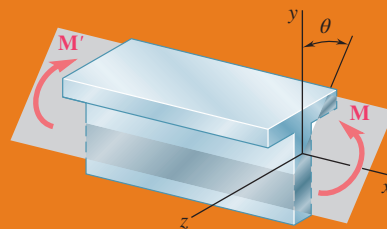


Fig. 4.72 Unsymmetric bending with bending moment not in a plane of symmetry.

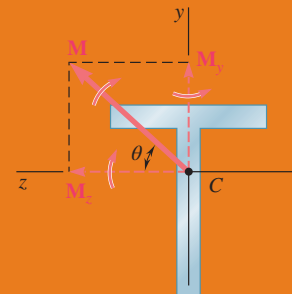


Fig. 4.73 Applied moment resolved into y and z components.

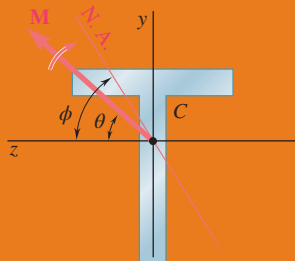


Fig. 4.74 Neutral axis for unsymmetric bending.

For the couple $\mathbf{M}$ shown in Fig. 4.74, the orientation of the neutral axis is defined by

$$\tan \phi = \frac{I_z}{I_y} \tan \theta \quad (4.57)$$

For the general case of *eccentric axial loading*, the load is replaced by a force-couple system located at the centroid. The stresses are superposed due to the centric load and the two component couples directed along the principal axes:

$$\sigma_x = \frac{P}{A} - \frac{M_z y}{I_z} + \frac{M_y z}{I_y} \quad (4.58)$$

In the analysis of stresses in *curved members* (Fig. 4.75), transverse sections remain plane when the member is subjected to bending. The *stresses do not vary linearly*, and the neutral surface does not pass through the centroid of the section. The distance R from the center of curvature of the member to the neutral surface is

$$R = \frac{A}{\int \frac{dA}{r}} \quad (4.66)$$

where A is the area of the cross section. The normal stress at a distance y from the neutral surface is

$$\sigma_x = -\frac{My}{Ae(R - y)} \quad (4.70)$$

where M is the bending moment and e is the distance from the centroid of the section to the neutral surface.

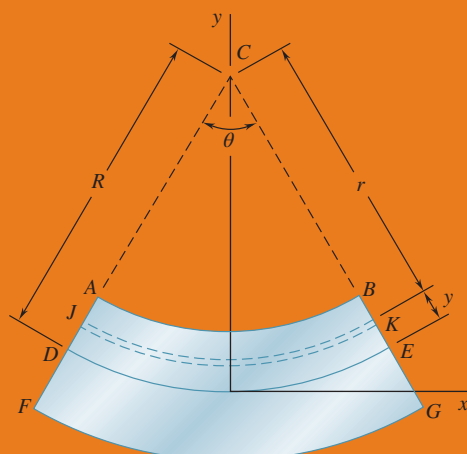


Fig. 4.75 Curved member geometry.